

INTERNSHIP FINAL PAPER

Internship Final Paper

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Old Dominion University - Spring 2026

CYSE 368: Cybersecurity Internship

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Twigs Naturals

April 19, 2026

Table of Contents

Introduction and Internship Overview	4
Management and Organizational Environment	6
Work Duties, Responsibilities, and Projects	7
Application of Cybersecurity Skills and Knowledge.....	11
Academic Preparation and Curriculum Relevance	14
Evaluation of Internship Learning Objectives	16
Motivating Aspects of the Internship	18
Discouraging Aspects of the Internship	20
Challenging Aspects of the Internship	21
Recommendations for Future Interns.....	23
Conclusion and Future Implications	24
Appendix A - Work Samples.....	27
Appendix A1 - WordPress Spam Mitigation Project	27
Figure A1.1 - “My Account” Page Before Spam Mitigation Implementation.....	27
Figure A1.2 - “My Account” Page After Spam Mitigation Implementation	28
Figure A1.3 - Checkout Page Preserving Login and Purchase Functionality	28
Figure A1.4 - Controlled Account Creation During Checkout	29
Figure A1.5 - Example of Spam User Accounts in WordPress “Users” List.....	30
Appendix A2 - Email Security Implementation Plan Project	31

INTERNSHIP FINAL PAPER

Figure A2.1 - SPF Record Validation Using MxToolbox	31
Figure A2.2 - Email Authentication Results (SPF, DKIM, and DMARC PASS).....	32
Appendix A3 - AppSheet Access Control Framework Project	33
Figure A3.1 - Unauthorized Access Handling (Application-Level Enforcement)....	33
Figure A3.2 - AppSheet Access Denied (System-Level Enforcement)	34
Figure A3.3 - Standard User Access (Partial Application Functionality)	35
Figure A3.4 - Administrator Access (Full Data and Administrative Features)	35
Figure A3.5 - Standard User Access (Restricted Functionality)	36
Figure A3.6 - Action-Level Security Enforcement (Delete Permissions).....	37

INTERNSHIP FINAL PAPER

Introduction and Internship Overview

As I planned for the final semester of my Bachelor of Science in Cybersecurity at Old Dominion University (ODU), I realized that the completion of an internship was both a requirement and a chance to gain real-world experience. The idea of applying my academic knowledge in a professional setting was especially appealing, particularly in areas related to governance, risk, and compliance (GRC), which has become a key area of interest for me. When I accepted an offer from Twigs Naturals as a cybersecurity technician intern, I realized that it would allow me to gain experience while working in a fully remote environment for the first time, which was a unique addition to my professional development.

Twigs Naturals is a Caribbean-based tea company based in Arima, Trinidad and Tobago. The organization produces and distributes herbal teas while growing its digital presence through its e-commerce website and various cloud platforms. The company uses a combination of tools to support its operations, including Google Workspace for internal communication and collaboration, WordPress with WooCommerce for its website and online store, HubSpot for customer relationship management (CRM), and the eFense vCISO Portal for policy management, user training, and compliance tracking. These systems support the company's business processes while introducing cybersecurity considerations that need to be managed effectively.

At the beginning of my internship, I attended a remote orientation session on Google Meet alongside another IT intern. During this session, I was introduced to the company's structure, its reliance on cloud-based tools, and the expectations of my role. One of my early observations was that the organization works in a collaborative and flexible environment across multiple platforms, including WhatsApp and Google Workspace (for both Gmail and secure

INTERNSHIP FINAL PAPER

sharing of documents). This was very different than my previous work experiences in more traditional, in-person environments, and it required me to quickly adapt to remote communication and self-directed task management.

The learning objectives were defined in my Memorandum of Agreement (MOA) and served as the foundation of my experience. These included developing experience in cybersecurity monitoring and analysis, applying network security principles to assess risk and improve security posture, supporting IT operations like onboarding and access management, and improving communication skills through documentation and collaboration. While these objectives initially seemed to align better with more technical roles, my internship demonstrated how these goals can also be achieved through GRC work that focuses on policy management, access control, and organizational processes.

During the earlier stages of my internship, my responsibilities mainly focused on supporting GRC efforts. This included helping with user management and training within the eFense vCISO Portal, and coordinating the intake and organization of standard operating procedure (SOP) documents. These initial tasks provided insight into how cybersecurity consists of more than purely technical controls; it also has a role in supporting business operations. As I progressed, my responsibilities expanded into more complex projects that required analyzing security gaps, proposing solutions, and contributing to long-term cybersecurity planning.

This experience provided an opportunity to explore how cybersecurity functions within a real-world business environment, particularly through the lens of GRC. The experience allowed me to apply concepts learned throughout my academic career, and it also challenged me to adapt those concepts to practical scenarios involving communication, documentation, and

INTERNSHIP FINAL PAPER

organizational coordination. This paper will examine my internship in detail, including my work, the skills I applied and developed, and how this experience has shaped my professional goals.

Management and Organizational Environment

The management environment at Twigs Naturals has a flexible, collaborative, and largely remote structure. As a cybersecurity technician intern, I reported directly to the Chief Executive Officer (CEO), which provided a unique level of visibility into operations and decision-making processes. This direct reporting structure made communication and feedback more efficient, but it also required a high level of independence, accountability, and professionalism in creating and managing tasks for my assigned projects.

Supervision throughout the internship was conducted through both synchronous and asynchronous communication methods. WhatsApp served as a communication platform for day-to-day coordination, quick updates, and informal discussions with team members. Google Meet was used for scheduled meetings, including weekly meetings and additional team discussions as necessary. Email communication was used heavily for formal updates and documentation sharing, especially when communicating progress to either the CEO or the Executive Assistant. This multi-channel approach required adaptability, as different types of information were shared across platforms depending on urgency and level of formality.

The organizational structure includes a combination of full-time staff and interns working across IT, HR, production, supply chain, finance, and marketing. Collaboration across all departments was an important part of my role, especially when coordinating tasks such as collecting policy documents, tracking compliance, and supporting system-related initiatives. In many cases, my role required me to interact with individuals who did not have a technical

INTERNSHIP FINAL PAPER

background, which emphasized the importance of communicating clearly and effectively. Task management was facilitated through a cloud-based Task Tracker system built in AppSheet. This allowed for the assignment, tracking, and prioritization of tasks across all team members, which provided visibility into the status and deadlines of tasks.

Compared to my unrelated work experience across two other in-person jobs since 2020, the management environment at Twigs Naturals is more flexible and decentralized. In both of my other jobs, tasks have been assigned in a direct and immediate manner within a physical workspace, and communication has been face-to-face. In contrast, Twigs Naturals operates in a remote environment for many employees and interns, with communication being distributed across multiple platforms, and tasks being more self-directed. This distinction emphasized the importance of time management, initiative, and proactive communication.

The management environment at Twigs Naturals effectively supported my internship experience by providing opportunities for both independent work and cross-functional collaboration. While the remote structure created certain challenges, including coordinating across multiple platforms and managing tasks without constant supervision, it also encouraged developing professional skills that are important in cybersecurity roles. These include the ability to communicate clearly in digital environments, manage responsibilities independently, and contribute to team objectives within a flexible organizational structure.

Work Duties, Responsibilities, and Projects

Throughout my internship, my responsibilities evolved from supporting foundational GRC activities to contributing to more structured security planning and analysis efforts. My work spanned multiple areas of cybersecurity, including access control management, policy

INTERNSHIP FINAL PAPER

coordination, risk mitigation, and security strategy development. Each of these responsibilities helped improve the company's overall cybersecurity posture by ensuring that systems, users, and processes aligned with security best practices.

One of my responsibilities involved supporting GRC efforts through the eFense vCISO Portal, which serves as the organization's centralized platform for policy management, user training, and compliance tracking. Within this system, I helped manage user accounts, correct user information, remove inactive users, and monitor the completion of required security training and policy acknowledgements. This work emphasized the importance of maintaining accurate access control and ensuring that all users are following organizational security policies. By tracking compliance and following up with employees and interns who had not completed their required tasks, I helped reinforce a culture of security awareness within the company.

In addition to managing compliance activities, I helped coordinate the intake and organization of SOPs and policy documents. This involved communicating with various departments to request documentation, confirming proper storage locations, and tracking the status of each document as I gathered signatures from the original author, the appropriate reviewer, and the authorizer (the CEO). This process emphasized the importance of documentation as a form of security control, as SOPs and other policies define how systems and processes should be used safely throughout the organization. My participation in these efforts required strong organizational skills and consistent communication to ensure that documentation efforts remained on track.

As my responsibilities expanded, I developed an Access Control Matrix and an accompanying application inventory to support onboarding and offboarding processes. The Access Control Matrix mapped organizational roles to the systems required for their

INTERNSHIP FINAL PAPER

responsibilities, which helped define the appropriate permission levels based on job function.

The application inventory documented important information about each system, including its business function, owner, data sensitivity, and authentication method. This project reinforced key cybersecurity concepts such as role-based access control (RBAC) and the principle of least privilege, which are both important for decreasing risk and preventing unauthorized access to sensitive information.

Another project involved addressing an issue related to the creation of spam accounts on the company's WordPress website. To mitigate this risk, I was tasked with developing a method of procedure (MOP) document that outlined potential solutions and evaluated their impact on system functionality. After reviewing multiple options, I proposed a configuration-based solution that involved modifying WooCommerce account settings to disable account registration from the login page while allowing account creation during the checkout process. This approach decreased the attack surface without introducing additional complexity or requiring the installation and configuration of new plugins. After implementing the changes, I validated the functionality of the website to ensure that legitimate users could still create accounts and complete transactions successfully. This project emphasized how even small configuration changes can require extensive testing across multiple website functions. However, it also demonstrated how targeted configuration changes can improve both security and operational efficiency.

In the later stages of my internship, I contributed to the development of an Email Security Enhancement Plan designed to implement authentication mechanisms such as SPF, DKIM, and DMARC. This project required analyzing the organization's existing email infrastructure, identifying gaps in authentication controls, and assessing the risks associated with those gaps,

INTERNSHIP FINAL PAPER

such as spoofing and phishing attacks. I also evaluated how various systems, including the company's email provider, website, and customer communication platforms, interacted within the organization's domain. Based on these findings, I developed a phased implementation strategy that included monitoring and gradual enforcement to minimize disruption to business operations. This project improved my understanding of how policy-driven configurations can be used to secure communication channels and defend organizational assets.

I also collaborated on a project involving the documentation and evaluation of access control mechanisms within applications built using AppSheet. This work focused on defining user roles, permissions, and access logic within the applications, with the goal of enabling scalable and controlled access to systems. I assisted in documenting how access should be managed within the applications themselves, rather than relying exclusively on external access restrictions. In other words, for some of these applications, access control was enforced within the applications themselves rather than relying solely on the underlying AppSheet infrastructure. Additionally, I participated in testing these controls by using both authorized and unauthorized user accounts to verify that permissions were correctly enforced. This experience emphasized the importance of aligning implemented controls with documented security policies and validating their effectiveness through testing.

Another area of contribution involved supporting the development of an archival and data governance framework, which served as a part of a broader initiative (across multiple departments) to improve customer service processes. In my role within this assigned cross-functional team, I helped define structured approaches for organizing and retaining data, including proposing naming conventions, access control guidelines, and retention timelines. This work emphasized the importance of data integrity and lifecycle management in cybersecurity, as

INTERNSHIP FINAL PAPER

improperly managed data can create both security risks and operational inefficiencies. By contributing to this initiative, I learned how data governance supports both compliance and overall organizational effectiveness.

Collectively, my responsibilities across numerous projects demonstrated how cybersecurity extends beyond technical implementations to include policy management, access control, and organizational processes. Each project helped improve the organization's security posture by addressing risks, improving system configurations, and encouraging accountability among employees and interns. Each of these experiences provided valuable insight into how cybersecurity practices can be applied within a real-world business environment, and how various aspects of security work together to defend systems and data. Detailed examples of selected work are included in Appendix A.

Application of Cybersecurity Skills and Knowledge

Throughout my internship, I applied a combination of foundational knowledge gained from academic coursework, and new skills developed through hands-on experience in this professional environment. While my coursework provided an understanding of technical concepts such as network security, system protection, and risk identification, my internship allowed me to apply these concepts in practical scenarios, especially in the context of GRC. This experience emphasized the idea that cybersecurity is not limited to technical defenses; it also includes policy enforcement, access control, and organizational coordination.

Before beginning my internship, I had developed a foundational understanding of cybersecurity principles through courses such as Cybersecurity Fundamentals, Introduction to Networking & Security, and Cybersecurity Strategy & Policy. These courses introduced key

INTERNSHIP FINAL PAPER

concepts such as defending systems and data, identifying vulnerabilities, and implementing security controls. One of the core learning outcomes of my degree is the ability to manipulate and protect computer systems, networks, and online data from attacks. Within my internship, I applied this outcome across multiple projects, including the development of an Access Control Matrix for defining system permissions based on user roles, and supporting policy enforcement within the eFense platform. Both projects demonstrated how structured access management reduces the risk of unauthorized access and defends organizational assets.

Another learning outcome for my degree is the ability to apply troubleshooting practices and identify potential security risks. This was directly involved in my internship when addressing the WordPress spam account issue and analyzing gaps in the company's email security configuration. In the case of WordPress, I evaluated how the account registration process created an attack surface that allowed numerous spam accounts to be created. By proposing and implementing a configuration change that restricted account creation to the checkout screen, I resolved this vulnerability while maintaining system usability. Similarly, when developing the Email Security Enhancement Plan, I identified the absence or misconfiguration of authentication mechanisms, including SPF, DKIM, and DMARC, and determined the associated risks, such as phishing and spoofing attacks. These experiences demonstrated how identifying and addressing security gaps is an important part of maintaining a secure environment.

While my degree also includes a learning outcome related to examining and collecting forensic evidence, this was less applicable to the responsibilities of this internship. However, the analytical and detail-oriented thinking that's required for forensic work was still useful for evaluating system configurations, reviewing documentation, and validating the effectiveness of implemented controls. For example, when I tested the access control mechanisms in company

INTERNSHIP FINAL PAPER

software within AppSheet, I approached the process by verifying expected behaviors for both authorized and unauthorized users. This approach reflects the attention to detail and validation practices that are used in forensic analysis, even though formal forensic investigations were not part of my internship.

In addition to applying existing knowledge, my internship required me to develop skills that weren't covered in my coursework. A substantial portion of my work involved GRC processes, such as policy management, compliance tracking, and documentation coordination. While courses such as Cybersecurity Strategy & Policy and Cybersecurity Ethics introduced high-level concepts related to governance and compliance, my internship provided hands-on experience in applying these concepts in a real organization. For example, coordinating the intake and organization of SOP documents demonstrated how policies serve as enforceable security controls, and monitoring training completion within the eFense platform emphasized the importance of user accountability in maintaining compliance.

Another area of skill development included professional communication. By drafting documentation, coordinating with members of various departments, and reporting progress to leadership, I improved my ability to communicate cybersecurity concepts clearly to both technical and non-technical audiences. This aligns with some of my other degree requirements, including Written Communication I (College Composition I) & II (College Composition II), and the Cybersecurity Writing Intensive Course (Cyber Strategy & Policy), which collectively emphasize the importance of communicating clearly and effectively in professional settings. My internship required me to apply these communication skills in real-world scenarios, especially when explaining security requirements, requesting information, and ensuring that tasks were accurately completed.

INTERNSHIP FINAL PAPER

This internship allowed me to connect academic knowledge with practical application. By applying cybersecurity principles in areas like access control, risk identification, and policy enforcement, I was able to realize how theoretical concepts can be implemented in real-world environments. At the same time, I developed new skills in GRC processes, communication, and system analysis that expanded my understanding of cybersecurity beyond only technical domains. This combination of previous knowledge and new experience has improved my ability to approach cybersecurity challenges thoroughly and practically.

Academic Preparation and Curriculum Relevance

My academic experience provided a strong foundation that supported my ability to contribute effectively during my internship. Through a combination of technical coursework, policy-focused classes, and writing-intensive experiences, I developed an understanding of cybersecurity principles that I could apply in a real-world environment. At the same time, my internship revealed areas where academic preparation alone wasn't sufficient, such as GRC processes and cross-functional communication within an organization.

Several courses in the degree program directly contributed to internship readiness. Courses like Cybersecurity Fundamentals and Introduction to Cybersecurity introduced core concepts like system protection, threat awareness, and security controls. These topics helped me understand the importance of defending systems and data from unauthorized access, which was applicable when working on projects like the Access Control Matrix and supporting policy enforcement within the eFense platform. Similarly, Introduction to Networking & Security offered knowledge on how systems communicate, and how network-based vulnerabilities can arise, which supported my ability to analyze risks related to email authentication mechanisms.

INTERNSHIP FINAL PAPER

In addition to technical courses, policy and strategy-oriented coursework played an important role in preparing me for my internship. Cybersecurity Strategy & Policy and Cyber Law introduced concepts related to governance, regulatory considerations, and organizational responsibility in maintaining security. While these courses mainly focused on theoretical frameworks, they provided context for understanding the importance of policies and compliance efforts within organizations. In this internship, I applied these concepts when coordinating SOP documentation, tracking policy acknowledgments, and supporting compliance initiatives through the eFense platform. This connection between academic theory and practical application emphasized the relevance of policy-driven approaches to cybersecurity.

Another part of my academic preparation was the focus on professional communication. Courses like Written Communication I & II, along with the Cybersecurity Writing Intensive Course, helped me develop the ability to clearly convey complicated ideas and concepts. These skills were important in my internship, as I was required to communicate with team members across various departments, request information, document processes, and report progress to leadership. The ability to write clearly and organize information effectively was especially valuable when working on structured tasks like SOP coordination and drafting documentation.

While my coursework established a strong theoretical foundation, my internship revealed areas where academic preparation was limited. One of the largest gaps was in hands-on experience with GRC processes. Even though I had been introduced to high-level concepts related to governance and policy, I hadn't previously worked with real-world systems like eFense vCISO Portal, or participated in managing policy lifecycles or compliance tracking. Furthermore, my coursework didn't fully prepare me for the level of cross-functional collaboration required in

INTERNSHIP FINAL PAPER

a professional environment, especially when communicating cybersecurity requirements to a non-technical audience.

My internship also required me to use tools and platforms that weren't covered in my degree program, such as cloud-based collaboration tools, CRM systems, and workflow management platforms. Learning to navigate these systems while maintaining a security-focused mindset required adaptability and independent learning. This internship demonstrated that while previous coursework provides essential knowledge, the ability to apply that knowledge depends on practical experience in real-world environments.

My academic preparation at ODU played a significant role in enabling my success during the internship by providing foundational knowledge, analytical skills, and communication abilities. However, the internship itself was important for forming the connection between theory and practice, especially for areas like GRC processes, organizational communication, and the use of real-world cybersecurity tools. Collectively, these experiences have resulted in a more thorough understanding of cybersecurity and have prepared me even more for future professional opportunities in the field.

Evaluation of Internship Learning Objectives

At the beginning of my internship, I created a few learning objectives through my MOA, which were intended to guide my professional development and provide structure to my internship. These objectives focused on developing hands-on cybersecurity experience, applying network security principles, supporting IT operations, and improving my communication skills. Throughout my internship, I was able to make progress toward each objective, although some were achieved in ways that differed from my initial expectations.

INTERNSHIP FINAL PAPER

The first objective was to develop hands-on experience in cybersecurity monitoring and analysis. While my internship didn't involve traditional network monitoring tools or incident response activities, I gained experience in analyzing organizational systems and identifying security risks. For example, when developing the Email Security Enhancement Plan, I evaluated the company's existing email infrastructure and identified the absence of SPF, DKIM, and DMARC authentication mechanisms. This required analyzing how different systems interacted within the company's domain and assessing the potential impact of these risks. Furthermore, my work in monitoring policy compliance and user training completion within the eFense platform improved my understanding of how ongoing monitoring supports an organization's security posture.

The second objective was to apply network security principles by identifying vulnerabilities, assessing compliance, and developing recommendations to improve security. This objective was reflected in several projects, especially the WordPress spam mitigation effort and the Access Control Matrix development. In WordPress, I identified a vulnerability in the account registration process that allowed spam accounts to be created rapidly, and I proposed a configuration-based solution to decrease this risk. Similarly, the Access Control Matrix project involved applying principles such as RBAC and least privilege to define appropriate system permissions for different user roles. These experiences demonstrated how theoretical security concepts can be applied to real-world systems to improve overall security.

The third objective was to improve communication skills through documentation and collaboration. This objective was achieved in nearly every aspect of my internship. I communicated often with team members across multiple departments to request information, coordinate tasks, and provide updates on project progress. Furthermore, I contributed to the

INTERNSHIP FINAL PAPER

creation and organization of documentation, such as SOP tracking, procedural planning, and structured frameworks for data management. These experiences helped me improve my ability to communicate technical concepts clearly and effectively, especially when working with employees and interns who did not have a technical background.

The internship successfully fulfilled the learning objectives outlined in my MOA. While some objectives were achieved through GRC-focused activities instead of more traditional technical roles, the experience provided insight into how cybersecurity works in an organization. By applying security principles, supporting operational processes, and improving my communication skills, I was able to meet the goals of the internship and gain a more thorough understanding of what it's like to work in a cybersecurity role.

Motivating Aspects of the Internship

One of the most motivating parts of my internship was being able to contribute to real-world cybersecurity efforts that directly impacted the organization. Unlike academic assignments, the work I performed at Twigs Naturals involved systems, processes, and policies that were actively used. Knowing that my contributions, such as improving access control structures, supporting policy compliance, and helping mitigate security risks, were meaningful to the company, which provided a sense of purpose and motivation throughout the internship.

Another motivating factor was having the chance to see the results of my work. For example, when addressing the spam account creation issue, I proposed a solution, implemented configuration changes, and validated that the issue had been mitigated. Being able to observe the outcome of my work reinforced my understanding of cybersecurity concepts and made the experience more engaging. Similarly, the Email Security Enhancement Plan allowed me to

INTERNSHIP FINAL PAPER

analyze a real-world problem and design a solution, which improved my confidence in applying cybersecurity knowledge in a professional setting.

The level of responsibility and independence I was given during the internship was also motivating. Reporting directly to the CEO and being trusted with tasks like developing documentation, analyzing systems, and proposing solutions encouraged me to take ownership of my work. This level of trust pushed me to a high standard of professionalism and accuracy, as my work had the potential to influence organizational decisions and processes.

Furthermore, the collaborative nature of the internship also contributed to my motivation. Working with individuals across multiple departments allowed me to gain a better understanding of how cybersecurity supports various aspects of business. Communicating with team members, gathering information, and coordinating tasks provided insight into how security efforts depend on cooperation between technical and non-technical roles. This aspect of my internship emphasized the importance of teamwork in achieving organizational goals.

Lastly, my growing interest in GRC as a potential career path was reinforced throughout my internship. As I became more involved in projects related to policy management, access control, and risk assessment, I found the work to be rather meaningful. My internship helped me recognize that cybersecurity is not limited to technical implementation; it also involves structured processes and decision-making that contribute to long-term organizational security. This realization was motivating because it caused my internship to align more closely with my personal strengths in organization, communication, and attention to detail.

The combination of real-world impact, observable results, professional responsibility, collaboration, and alignment with my career interests made this internship highly motivating.

INTERNSHIP FINAL PAPER

These factors helped me remain focused on my work and continue developing my cybersecurity skills.

Discouraging Aspects of the Internship

While my internship was a positive experience overall, there were certain aspects that I found discouraging at times. One of the main challenges was having to rely on communication and coordination across multiple departments, especially in a fully remote environment. Many responsibilities, including SOP & policy coordination and compliance tracking, depended on responses from others, which was not always efficient. In some cases, communication delays or incomplete responses slowed down progress, which slowed my ability to progress with tasks.

Another discouraging part was the limited amount of control I had over some systems and processes. As an intern, my role typically involved identifying issues, proposing solutions, and implementing approved solutions, but final decisions and approvals were dependent on leadership or other stakeholders. While this is a normal part of professional settings, it was sometimes frustrating when progress depended on factors outside of my control, especially when I had already finished my portion of a task and was waiting for feedback or approval.

The remote nature of my internship also introduced some difficulties, despite being beneficial in some ways. Without the ability to communicate face-to-face, it was sometimes challenging to clarify requirements, confirm expectations, or ensure that everyone was aligned. Communication across multiple platforms, including WhatsApp, Gmail, Google Drive, and the company's Task Tracker software, often led to fragmented information, which was tedious to track and organize effectively.

INTERNSHIP FINAL PAPER

Furthermore, some of my responsibilities involved repetitive or administrative tasks, such as monitoring training completion, tracking policy acknowledgments, and following up with individuals who had not completed their tasks within the eFense platform. While these tasks were necessary for ensuring compliance and supporting organizational security, they were not always as engaging as more analytical or project-based work. Regardless, these responsibilities still impacted my understanding of how consistency and attention to detail are important in cybersecurity operations.

Despite these discouraging aspects, each challenge also served as an opportunity to develop greater patience, adaptability, and problem-solving skills. Learning how to navigate communication delays, work within organizational constraints, and remain productive in a remote environment helped me grow professionally. These experiences emphasized the importance of being flexible and persistent when working in cybersecurity, especially when multiple stakeholders and processes must be coordinated effectively.

Challenging Aspects of the Internship

One of the most challenging aspects of my internship was balancing security considerations with practical business needs. For example, many tasks I worked on required finding solutions to improve security without harming usability or business operations. This was particularly noticeable when addressing the spam account creation issue on the company's website, as it was important to reduce the attack surface while allowing legitimate customers to create accounts and complete transactions. This required me to carefully evaluate potential solutions and consider how each one would affect both security and user experience.

INTERNSHIP FINAL PAPER

Another challenge was understanding and working within GRC processes, which were not a primary focus of my academic coursework. While I had been exposed to high-level concepts related to policy and strategy, applying these concepts in a real organization required a deeper level of understanding. Tasks such as coordinating SOP documentation, tracking policy approvals, and ensuring compliance with organizational standards required attention to detail and consistency, along with the ability to interpret how policies translate into real practices. This learning curve made these responsibilities challenging, but it also significantly contributed to my professional development while I'm still attending ODU.

The complexity of working with multiple systems and platforms was also challenging. Throughout my internship, I interacted with various tools, such as the eFense vCISO Portal, WordPress with WooCommerce, Google Workspace, HubSpot, and AppSheet-based applications. Each system had its own functionality, configuration options, and security considerations. Understanding how these systems interact with each other, especially when analyzing email security or access control structures, required significant time and effort. Becoming familiar with these platforms while remaining focused on security best practices was an important but challenging part of my internship.

Furthermore, communicating cybersecurity concepts to individuals without a technical background was sometimes challenging. When coordinating tasks or requesting information from other departments, I needed to make sure that my communication was clear, concise, and understandable. This required me to adjust my language depending on the audience, which was a skill I had to develop throughout the internship. It was challenging to try to bridge the gap between technical and non-technical audiences, but it was necessary to ensure that security practices were actually understood and followed.

INTERNSHIP FINAL PAPER

Ultimately, the challenges I faced in my internship positively impacted my growth as a cybersecurity professional. By learning how to balance security and usability, navigate GRC processes, work across multiple systems, and communicate effectively with diverse audiences, I developed a deeper understanding of cybersecurity. These challenges emphasized the importance of adaptability, critical thinking, and continuous learning in cybersecurity.

Recommendations for Future Interns

Based on my internship experience, I would offer several recommendations for future interns who are preparing for a similar role. First, it is important to have a foundational understanding of cybersecurity concepts, especially in areas like access control, risk management, and basic network security. While advanced technical expertise may not be required, having a better understanding of core principles will make it easier to understand tasks and contribute effectively to assigned projects.

In addition to technical knowledge, strong organizational skills are a must. My responsibilities involved tracking multiple tasks, managing documentation, and coordinating with various departments. Being able to keep information organized, follow up on outstanding tasks, and maintain accurate records will help ensure work is completed accurately and efficiently. Attention to detail is especially important when working with policies, access permissions, and compliance-related activities.

Future interns should also be prepared to communicate consistently and clearly, especially in a remote internship. Since communication will likely take place across multiple platforms, including messaging applications, email, and virtual meetings, it's important to be proactive in asking questions, providing updates, and confirming expectations. Clear

INTERNSHIP FINAL PAPER

communication helps prevent misunderstandings and ensures that tasks are aligned with project goals.

Another recommendation is to be adaptable and open to learning new tools and systems. Interns may be required to work with platforms that they have not previously encountered, including security management systems, workflow applications, and more. Taking the initiative to learn how each system functions and how they relate to cybersecurity will improve both internship performance and the completion of the learning objectives.

Lastly, it's important for future interns to approach the experience with patience and a willingness to learn from challenges. While some tasks are more likely to involve delays, coordination with multiple stakeholders, or repetitive processes, they are still valuable experiences that are a part of understanding how cybersecurity operates within a real company. Maintaining a positive attitude and focusing on continuous improvement will help future interns make the most of their experience and develop skills that can be applied to future professional roles. By developing a strong foundation in cybersecurity concepts, maintaining organization and attention to detail, communicating effectively, adapting to new tools, and approaching challenges with a learning mindset, future interns can gain meaningful professional development.

Conclusion and Future Implications

My internship at Twigs Naturals served as a valuable component of my senior year in the Bachelor of Science in Cybersecurity program at ODU. Throughout my experience, I was able to apply cybersecurity knowledge in a real-world setting while also gaining new insights into how cybersecurity functions within an organization. By working on projects related to access control,

INTERNSHIP FINAL PAPER

policy management, compliance tracking, and security planning, I developed a better understanding of how GRC practices impact an organization's overall cybersecurity posture.

One of the most important takeaways from this experience is the realization that cybersecurity extends beyond technical tools and systems. While technical knowledge is important, my internship demonstrated that effective cybersecurity relies on organized processes, clear documentation, and consistent user accountability. My work with policy coordination, access management, and communication across departments emphasized the importance of aligning technical controls with organizational practices. This perspective has expanded my understanding of cybersecurity and emphasized the value of GRC-focused roles within the field.

This internship has influenced my senior year at ODU by shaping how I approach my final coursework and academic projects. With a greater appreciation for real-world applications, I am more prepared to connect theoretical concepts to practical scenarios, and to focus on areas that are relevant to professional environments. Furthermore, the experience has improved my confidence in my ability to meaningfully contribute to cybersecurity efforts, which will support me as I complete my final academic requirements.

Looking ahead, this internship has had a substantial impact on my future professional goals. Through this experience, I have developed a stronger interest in pursuing a career that incorporates both technical and governance aspects of cybersecurity. I have found that my strengths in organization, attention to detail, and communication align well with roles in GRC, policy management, and security operations. At the same time, I recognize the importance of continuously building my technical knowledge to complement my current skillset, so that I can remain adaptable in a field that is rapidly evolving.

INTERNSHIP FINAL PAPER

My internship at Twigs Naturals provided a meaningful learning experience that has prepared me to transition from academic study to professional practice. By combining theoretical knowledge with practical application, I have gained a more comprehensive understanding of cybersecurity and the role it has in supporting organizational success. This internship has not only fulfilled the requirements of my degree program, but has also impacted my career direction and long-term professional development within the cybersecurity field.

Appendix A - Work Samples

The following sections present selected work samples from my internship that demonstrate key cybersecurity concepts and contributions. Certain materials have been intentionally limited or modified to protect sensitive organizational and user information while still providing meaningful evidence of my work.

Appendix A1 - WordPress Spam Mitigation Project

This appendix presents selected evidence from the WordPress Spam Mitigation initiative, including system configurations before and after implementation, validation of functionality, and identification of spam account activity.

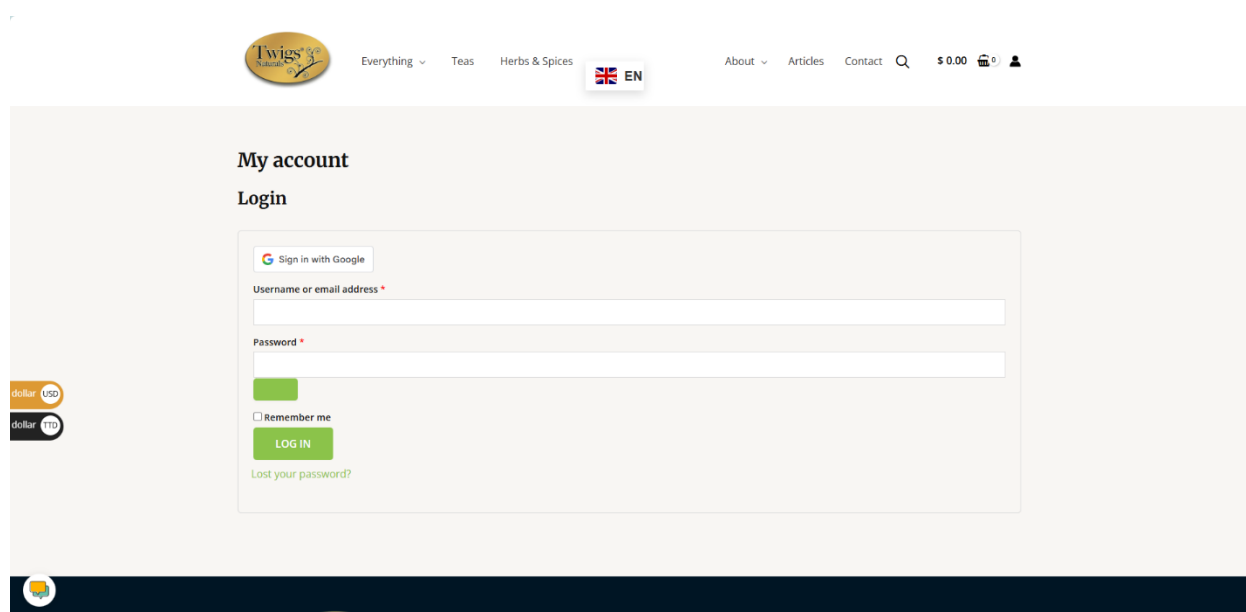
Figure A1.1 - “My Account” Page Before Spam Mitigation Implementation

The screenshot displays the 'My account' page of the Twigs website. The page features a header with the Twigs logo, navigation links (Everything, Teas, Herbs & Spices), a language selector (EN), and a shopping cart icon showing a total of \$0.00. The main content area is divided into two sections: 'Login' and 'Register'. The 'Login' section includes a 'Sign in with Google' button, input fields for 'Username or email address' and 'Password', a 'Remember me' checkbox, a green 'LOG IN' button, and a link for 'Lost your password?'. The 'Register' section includes input fields for 'Username' and 'Email address', a note about password reset links, a privacy policy link, and a green 'REGISTER' button. A chat bubble is visible in the bottom left corner, and a footer with 'Payment Information' is partially visible at the bottom right.

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This screenshot shows the Twigs Naturals “My Account” page prior to implementing security controls. Both login and account registration options were publicly accessible, enabling automated bots to create fraudulent user accounts through the WooCommerce registration form. This configuration enabled repeated automated registration attempts through a publicly exposed entry point.

Figure A1.2 - “My Account” Page After Spam Mitigation Implementation



This screenshot shows the Twigs Naturals “My Account” page after disabling public account registration. The registration form has been removed, while the login functionality remains available for existing users. This change eliminated the primary entry point used by automated bots to create spam accounts, while preserving user access through login and checkout-based account creation.

Figure A1.3 - Checkout Page Preserving Login and Purchase Functionality

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Checkout

☐ Returning customer? [Click here to login](#)

☐ Have a coupon? [Click here to enter your code](#)

Billing details

First name * Last name *

Company name (optional)

Country / Region *

Street address *

House number and street name

Apartment, suite, unit, etc. (optional)

Town / City *

Your order

Product	Subtotal
Bamboo Tea - Medium x 1	\$ 6.00
Subtotal	\$ 6.00
Total	\$ 6.00

☒ Direct bank transfer

Make your payment directly into our bank account.
Please use your Order ID as the payment reference.
Your order will not be shipped until the funds have cleared in our account.

This screenshot shows the Twigs Naturals checkout page after implementing spam mitigation controls. Existing users are still able to log in during checkout, and the purchasing process remains fully functional.

Figure A1.4 - Controlled Account Creation During Checkout

Email address *

☒ Create an account?

Account username *

☒ Ship to a different address?

First name * Last name *

Company name (optional)

Country / Region *

Street address *

House number and street name

Apartment, suite, unit, etc. (optional)

Town / City *

State / County *

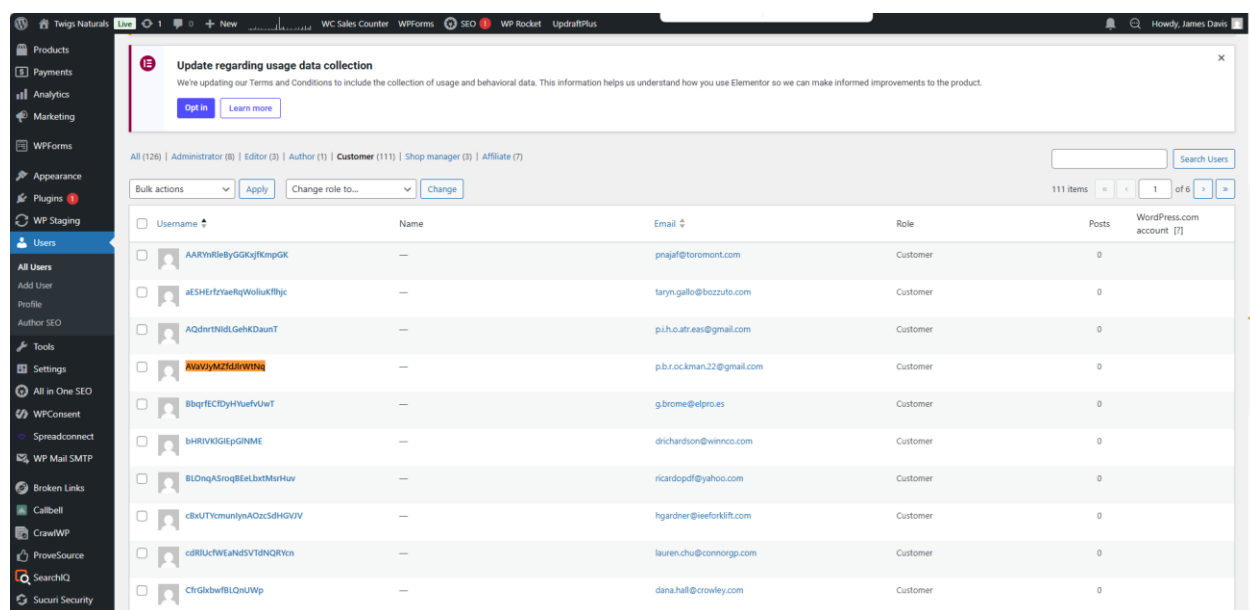
Postcode / ZIP *

Phone (optional)

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This screenshot shows the optional “Create an account?” feature during the checkout process. Account creation is only available when explicitly selected by the user, reducing exposure to automated abuse. By limiting account registration to the checkout process and requiring user interaction, the system prevents automated account creation while preserving a seamless onboarding experience for legitimate customers. This approach balances security and usability without introducing additional complexity.

Figure A1.5 - Example of Spam User Accounts in WordPress “Users” List



Note: Only spam user accounts are displayed in this screenshot; no legitimate accounts are shown to protect the privacy of real users.

This screenshot shows a portion of the WordPress “Users” list containing spam accounts. These accounts exhibit characteristics such as randomly generated usernames, missing profile information (no names), and no associated activity (0 posts). These patterns were used to identify

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and confirm fraudulent accounts created through automated registration. A total of 97 spam users were identified and removed as part of the mitigation process, improving the accuracy and integrity of the system's user data.

Appendix A2 - Email Security Implementation Plan Project

This appendix presents selected screenshots from the Email Security Implementation initiative, including both the validation and evidence of SPF, DKIM, and DMARC functionality. These tasks were achieved by using a validation tool and observing the authentication results of a test email that I sent from a company email account to my authorized internship email account.

Figure A2.1 - SPF Record Validation Using MxToolbox

SuperTool Beta9

twignaturals.com [SPF Record Lookup](#)

spf:twignaturals.com [Find Problems](#) [Solve Email Delivery Problems](#) [spf](#)

```
v=spf1 include:_spf.google.com include:spf.hubspotemail.net ~all
```

Prefix	Type	Value	PrefixDesc	Description
v		spf1		The SPF record version
+	include	_spf.google.com	Pass	The specified domain is searched for an 'allow'.
+	include	spf.hubspotemail.net	Pass	The specified domain is searched for an 'allow'.
~	all		SoftFail	Always matches. It goes at the end of your record.

Test	Result
DMARC Policy Not Enabled	It is recommended to use a quarantine or reject policy. To enable BIMI, it is required to have one of these at 100%. More Info
SPF Record Published	SPF Record found
SPF Record Deprecated	No deprecated records found
SPF Multiple Records	Less than two records found
SPF Contains characters after ALL	No items after 'ALL'.
SPF Syntax Check	The record is valid
SPF Included Lookups	Number of included lookups is OK
SPF Recursive Loop	Nor Recursive Loops on Includes
SPF Duplicate Include	No Duplicate Includes Found
SPF Type PTR Check	No type PTR found
SPF Void Lookups	Number of void lookups is OK
SPF MX Resource Records	Number of MX Resource Records is OK
SPF Record Null Value	No Null DNS Lookups found
DMARC Record Published	DMARC Record found

Your DNS hosting provider is "Squarespace" [Need Bulk Dns Provider Data?](#)

dns lookup dns check mx lookup dmARC lookup dns propagation

Reported by ns04.squarespacedns.com on 4/9/2026 at 4:47:40 PM (UTC -5) [just for you](#) [Transcript](#)

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Note: the “DMARC Policy Not Enabled” test result is intentional, because it was only feasible to enable SPF for HubSpot, unlike DKIM, which would require a feature that is only available to specific subscription tiers on HubSpot. Upon notifying company leadership about this limitation, I received verbal confirmation in a subsequent team meeting that the company will not be paying to upgrade its HubSpot subscription.

Accordingly, this resulted in SPF, DKIM, and DMARC being successfully implemented for Google Workspace (as proven in Figure A2.2), while only SPF is fully available for HubSpot, unlike DKIM (DMARC is partially enabled, but with “Monitoring;” not “Quarantine” or “Reject,” which are recommended).

This screenshot shows the successful deployment and validation of the SPF record for twigsnaturals.com using MxToolbox. The configuration authorizes Google Workspace and HubSpot as legitimate email sending sources. The SPF record defines which systems are permitted to send email on behalf of the domain, reducing the risk of email spoofing. Validation results confirm that the record is correctly configured, with no syntax errors, duplicate entries, or invalid lookups. This represents a foundational step in establishing secure email authentication.

Figure A2.2 - Email Authentication Results (SPF, DKIM, and DMARC PASS)

Original Message

Message ID	<CANWH12YK+qSKoVUT+0DcuZUwEwv3Vhi4R8+w6ss2QYHbnjZEg@mail.gmail.com>
Created at:	Thu, Apr 9, 2026 at 7:30 PM (Delivered after 13 seconds)
From:	Ajoupa Cloud Services <ajoupa@twigsnaturals.com>
To:	"jgdavis.tn@gmail.com" <jgdavis.tn@gmail.com>
Subject:	Email Authentication Test - Google Workspace
SPF:	PASS with IP 209.85.220.41 Learn more
DKIM:	'PASS' with domain twigsnaturals-com.20251104.gappssmtp.com Learn more
DMARC:	'PASS' Learn more

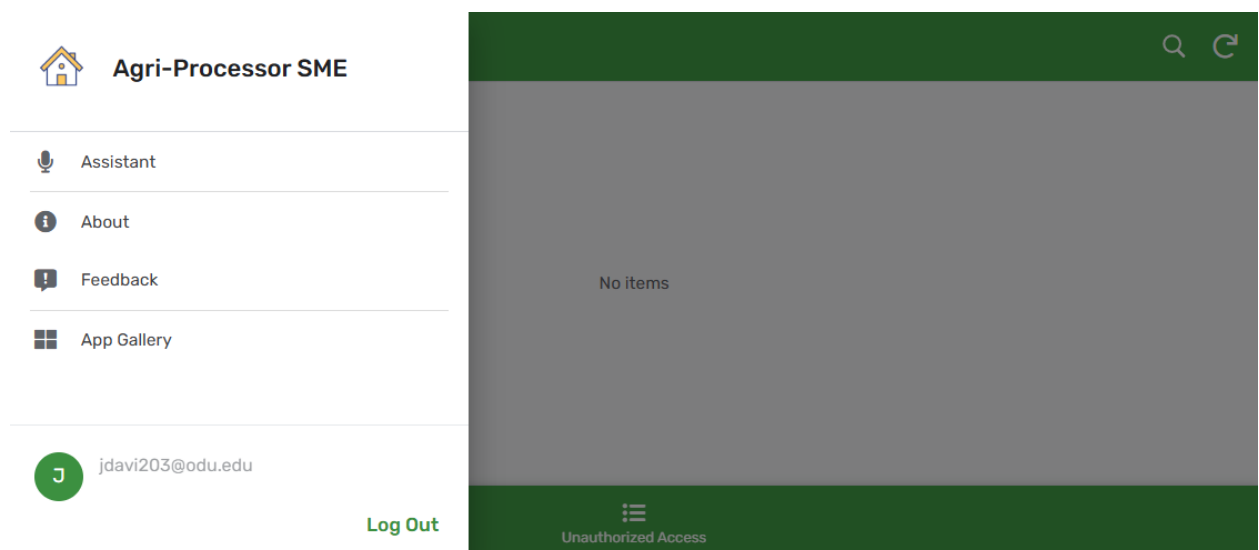
[Download Original](#) [Copy to clipboard](#)

This screenshot shows the authentication results of a test email sent from the twigsnaturals.com domain, confirming successful SPF, DKIM, and DMARC validation. This validation demonstrates that the implemented email security controls are functioning correctly. SPF confirms that the sending server is authorized, DKIM verifies message integrity and domain alignment, and DMARC confirms that both mechanisms pass under the defined policy. Together, these results indicate that the domain is protected against common email spoofing and impersonation attacks.

Appendix A3 - AppSheet Access Control Framework Project

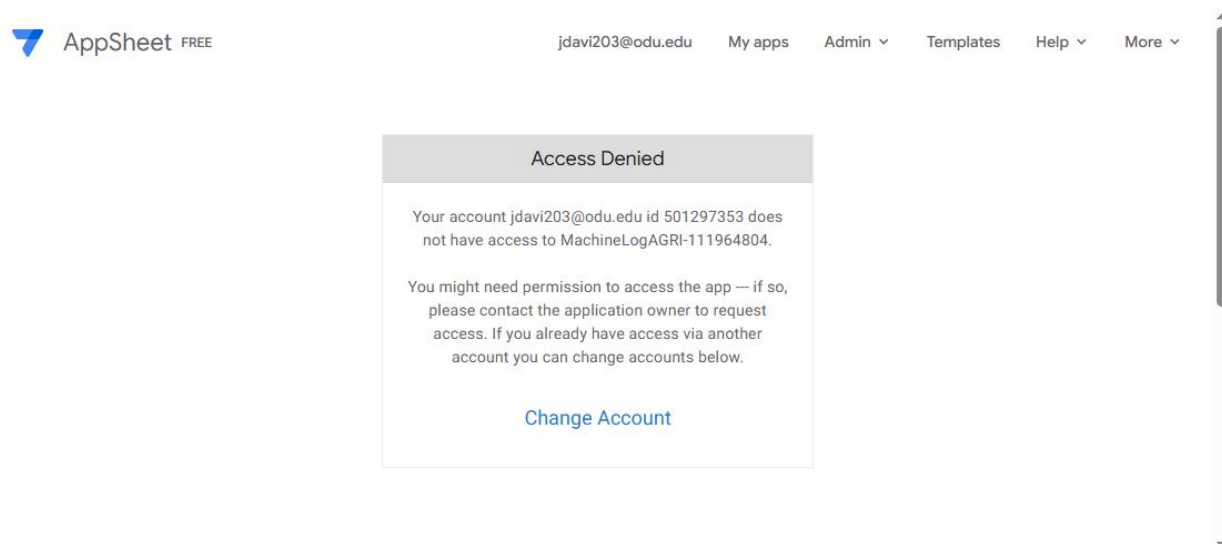
This appendix presents selected evidence from the AppSheet Access Control implementation, including role-based access scenarios, system and application-level enforcement, and validation of user permissions across multiple access levels.

Figure A3.1 - Unauthorized Access Handling (Application-Level Enforcement)



This screenshot shows a test unauthorized user (using my ODU email) accessing one of the AppSheet application interfaces that restricts access to data and functionality through application-level controls. Unauthorized users are prevented from viewing any meaningful data or interacting with system functionality, and are presented with an “Unauthorized Access” message at the bottom, along with “No items” displayed in the center. This demonstrates the implementation of a default-deny access control model, which I defined in the security framework for company software hosted on AppSheet.

Figure A3.2 - AppSheet Access Denied (System-Level Enforcement)

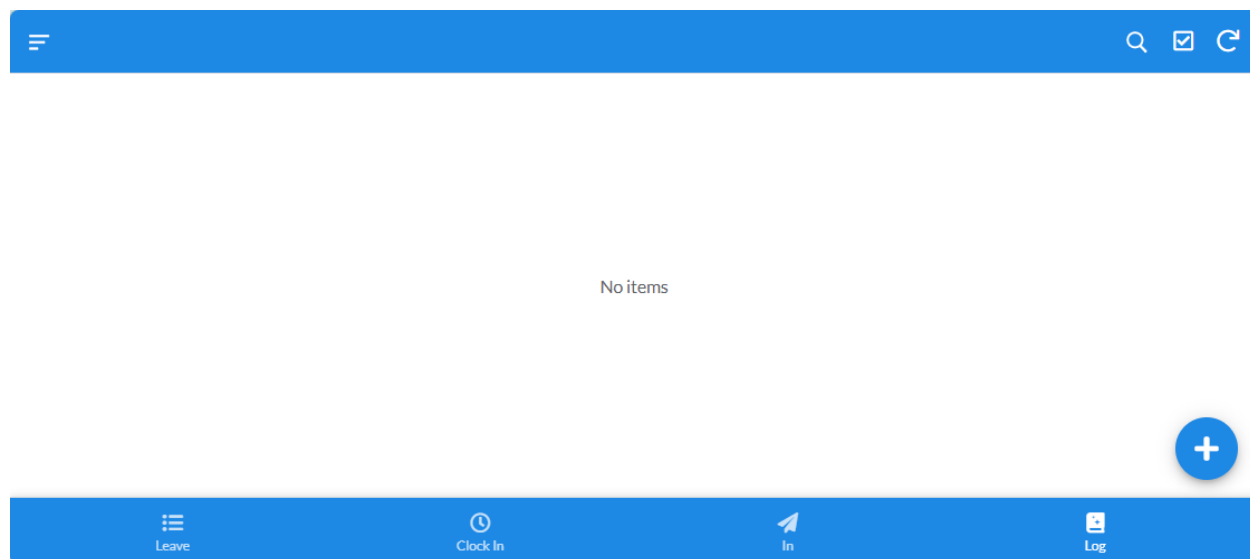


This screenshot shows AppSheet’s built-in access control preventing an unauthorized user from accessing the application entirely. In this scenario, access is restricted at the AppSheet platform level rather than within the application itself. The user is blocked from loading the application interface due to insufficient permissions in the app’s sharing configuration. This

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demonstrates an additional layer of access control, where unauthorized users are denied entry before any application logic is executed.

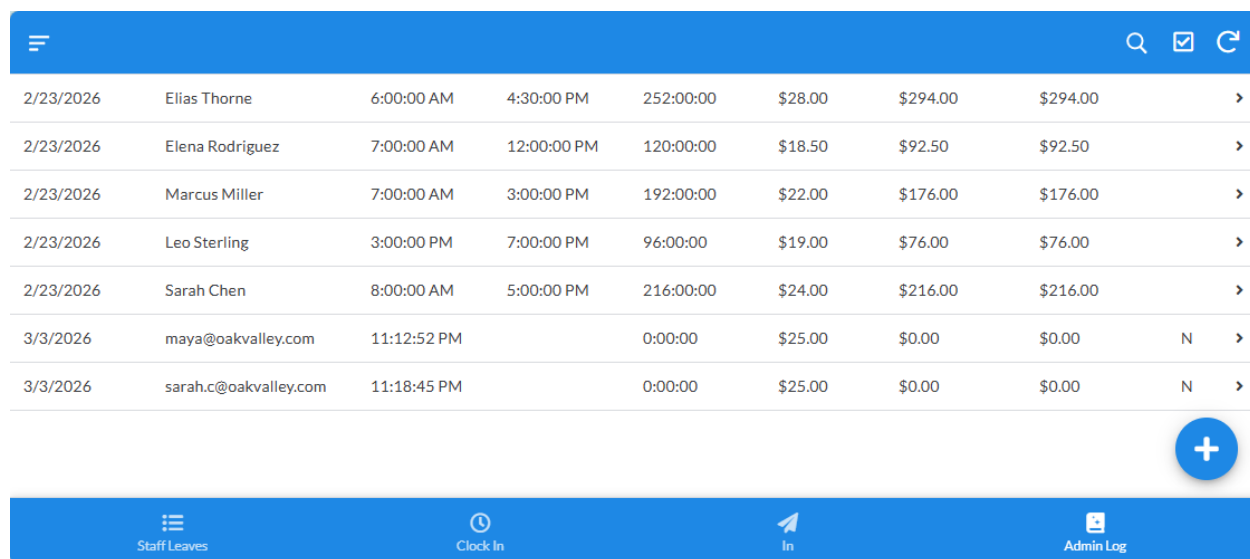
Figure A3.3 - Standard User Access (Partial Application Functionality)



This screenshot shows a standard user with partial access to application features and navigation options. Standard users are granted partial access to application functionality, including operational views and action controls. In this example, only “Log” is shown (instead of “Admin Log”), which is why this user cannot view any entries (for a Timesheet application) until they clock in or out (admins would be able to view entries for all employees). This reflects proper role-based access control, where standard user privileges only allow restricted interaction with application data and features.

Figure A3.4 - Administrator Access (Full Data and Administrative Features)

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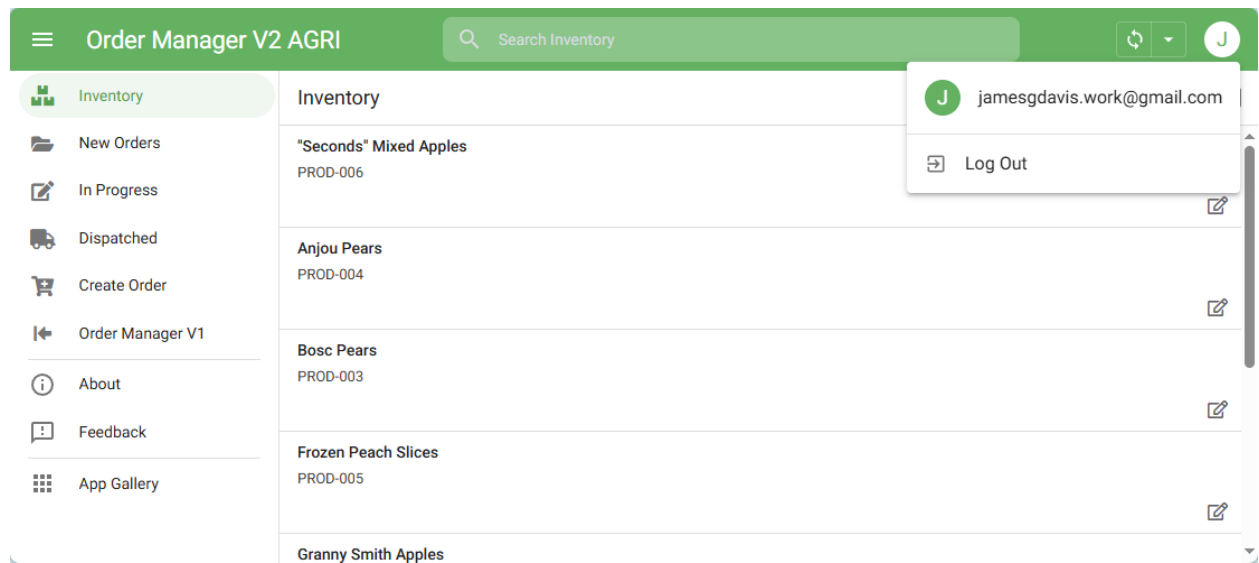


Date	Name	Start Time	End Time	Total Time	Payment 1	Payment 2	Payment 3	Action
2/23/2026	Elias Thorne	6:00:00 AM	4:30:00 PM	252:00:00	\$28.00	\$294.00	\$294.00	>
2/23/2026	Elena Rodriguez	7:00:00 AM	12:00:00 PM	120:00:00	\$18.50	\$92.50	\$92.50	>
2/23/2026	Marcus Miller	7:00:00 AM	3:00:00 PM	192:00:00	\$22.00	\$176.00	\$176.00	>
2/23/2026	Leo Sterling	3:00:00 PM	7:00:00 PM	96:00:00	\$19.00	\$76.00	\$76.00	>
2/23/2026	Sarah Chen	8:00:00 AM	5:00:00 PM	216:00:00	\$24.00	\$216.00	\$216.00	>
3/3/2026	maya@oakvalley.com	11:12:52 PM		0:00:00	\$25.00	\$0.00	\$0.00	N >
3/3/2026	sarah.c@oakvalley.com	11:18:45 PM		0:00:00	\$25.00	\$0.00	\$0.00	N >

Note: The users listed in this screenshot are fictional; they were only included for software development and testing purposes.

This screenshot shows an administrator-level user with full visibility of application data and access to administrative features, including the “Admin Log.” Administrator users are granted full access to application data and functionality. In this example, the user can view all records, including employee time and payment data. The presence of these features confirms proper role-based access control, where elevated privileges are restricted to authorized users only.

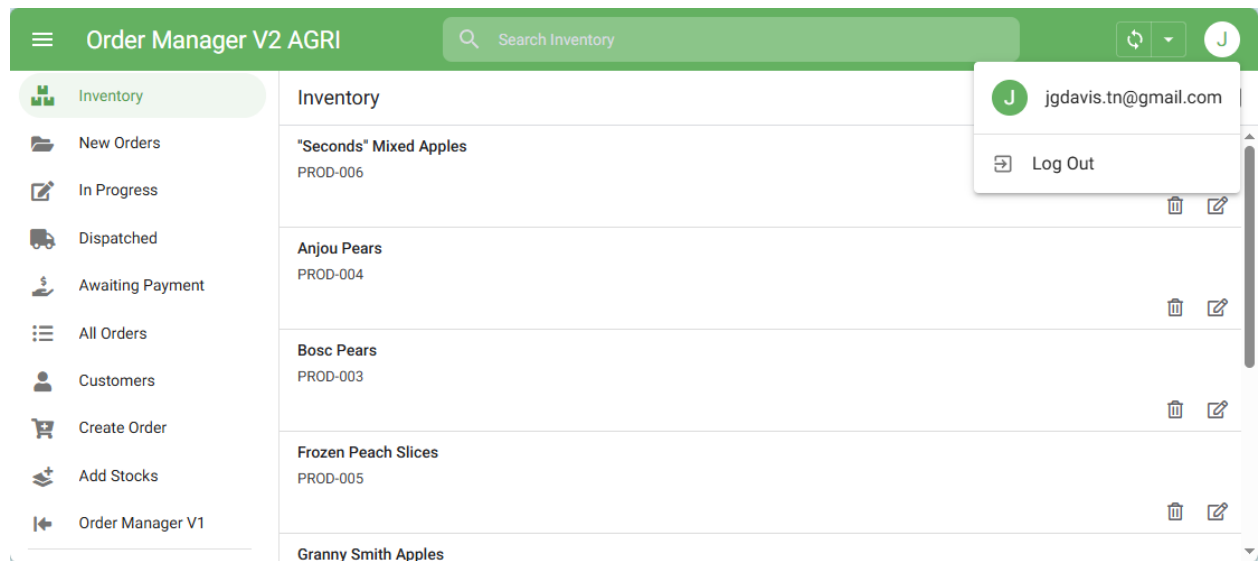
Figure A3.5 - Standard User Access (Restricted Functionality)



This screenshot shows a standard user with access to operational features while administrative functionality is restricted. Standard users are granted access to core operational features required for daily tasks, such as viewing inventory and managing orders. However, administrative functionality is restricted. In this example, the absence of delete (trash) icons demonstrates that elevated privileges are not available to standard users [the button would be located to the left of each edit (pencil) icon on the righthand side]. This confirms that role-based access control is properly enforced, limiting access based on user role.

Figure A3.6 - Action-Level Security Enforcement (Delete Permissions)

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This screenshot shows administrator-level access where destructive actions, such as deleting records, are available through visible delete icons. Action-level access control is enforced to restrict high-risk operations, such as deleting records, to administrator users only. In this example, delete (trash) icons are visible, indicating that the user has permission to perform destructive actions. Standard users are not granted access to these controls, ensuring that critical data cannot be modified or removed without appropriate authorization.